

Faithful attribution recovers the causal wiring, not the semantics: a ground-truth audit of interpretability on a fully known machine

Andreas Maier^{1*}, Stefan Bayer¹ and Patrick Krauss¹

^{1*}Pattern Recognition Lab, Friedrich-Alexander-Universität
Erlangen-Nürnberg, Erlangen, Germany.

*Corresponding author(s). E-mail(s): andreas.maier@fau.de;

Abstract

We ask whether a faithful explanation is already an understanding. Interpretability methods are judged by their faithfulness — whether they name the true causes — yet on real neural systems there is no ground truth against which to check them. We remove that excuse on a system whose ground truth is computable: a bit-exact, differentiable port of the Atari 2600, never built to be interpretable, in which the cause of any output follows from exact intervention. Scoring about 31 interpretability methods against this oracle, we find causal methods faithful, while popular gradient and correlational saliency methods are provably zero on discrete position outputs (activation patching **1.000** versus vanilla saliency **0.000**) yet score higher on human plausibility — a danger zone of conviction without correctness. Yet faithfulness is the smaller half of understanding: an exact method returns a causal-effect map or a data-flow graph, not the computation. A graph-correct circuit need not reproduce behavior (sufficiency **0.44**), and a feature matching a named variable can be causally inert (matched fraction **1.0**, faithfulness **0.04**). We name this measured residual the *semantic gap*, and stress that we *measure* it, not *close* it. As such, the approach offers a reusable ground-truth benchmark for explainable AI, mechanistic interpretability, AI safety, and software verification. We assume that it will help redirect the field toward recovering meaning, not merely confirming causes.

Keywords: interpretability, explainable AI, mechanistic interpretability, faithfulness, ground truth, causal attribution, semantic gap, Atari VCS

1 Introduction

When a method tells us what a system is doing, do we believe it because it is *true* or because it is *plausible*? And once we have established that its account is true, have we thereby *understood* the program, or only named its causes? These two questions sit at the centre of interpretability, and they are not the same question. A neural network now drives a car, reads a scan, and files a loan; before we trust such a decision we ask an explanation method why it was made. The explanation is load-bearing for safety and for science alike, yet on the systems we most want to explain we cannot check whether it is correct. Saliency maps for an Atari agent look convincing and shift with the action, but a convincing map is not a verified one [1]. The field has therefore converged on *faithfulness* — whether an explanation names the system’s *true* causes — as its quality criterion. Faithfulness is the right criterion. The difficulty is that on a real neural network there is nothing to measure it against.

This is the predicament Jonas and Kording made vivid by turning the neuroscientist’s toolkit on a microprocessor whose every transistor is known [2]. Lesions, tuning curves, and connectomics produced rich, suggestive structure — and none of it recovered how the chip actually computes. Their lesson generalizes: when the ground truth is unknown, a method that finds structure cannot be told apart from a method that finds the *right* structure. The same blind spot recurs across explainable AI, where attribution maps are validated against each other or against human intuition rather than against the system’s mechanism [3]. Lacking a reference, the community has quietly let faithfulness stand in for understanding, and cannot test whether the proxy holds. We remove that excuse.

The opportunity.

Paper 1 of this program built the missing reference: a bit-exact, differentiable port of the Atari 2600 Video Computer System (VCS) that reproduces a real, hand-authored 1977 machine to the byte and to the pixel [4, 5]. The VCS is software with a known specification, so the true cause of any output is not a matter of opinion — it is computable by exact intervention. We can clamp a RAM cell, replace a register, occlude a ROM byte, re-run the machine deterministically, and read off the change in the output. This gives us an *intervention oracle*: for every candidate cause of every output, the true causal effect, measured rather than estimated. Figure 1 lays out the platform and the oracle — the three faces of the system we can observe and intervene on (inputs, internal state, outputs) and the tiers of ground truth they afford. On this machine, for the first time on a real and complex artifact, an interpretability claim can simply be scored as right or wrong.

The move — confirm.

We turn the toolkit on this one system and score it. Across the neuroscience battery, the attribution and explainable-AI methods, and the mechanistic-interpretability toolkit — 31 methods aggregated from 257 per-game records — we place every method on two axes: faithfulness against the oracle, and a documented plausibility proxy that captures how convincing the method’s tradition tends to look. The result confirms

what the causal view predicts, now in numbers. Causal and intervention methods are faithful; popular gradient and correlational saliency methods are not. On discrete sprite-position outputs — where the naive gradient is provably zero, because position is set by strobe timing rather than by a value the gradient can flow through — the contrast is at its sharpest: activation patching reaches faithfulness 1.000, the oracle ceiling, while vanilla saliency collapses to 0.000, the naive-gradient floor, on every one of the six core games. Aggregated over that regime, causal and intervention methods average 0.412 faithfulness against 0.068 for the gradient and correlational family, a gap of 0.344. The popular method nonetheless carries the higher plausibility (0.9 versus 0.5): it looks more convincing precisely where it is least correct. This is the field’s danger zone — conviction without correctness — and Fig. 4 marks it out as a measured region of the plane, not a cautionary metaphor.

The deeper move — the boundary the leaderboard cannot cross.

Securing faithfulness, however, is the smaller half of the problem. This is the paper’s central claim, and it is where a ground-truth machine lets us say something the leaderboard alone cannot. With the same oracle we ask not only *which* methods are faithful but *what a faithful method delivers*, and the answer is a true causal-effect map or a true data-flow graph — not the variable’s meaning and not the routine’s algorithm. To make this precise we require that an explanation be right only if it is **F**aithful (its claimed causes are the true causes), **S**ufficient (it regenerates the behavior under held-out interventions), and **M**inimal at the algorithmic level (no spurious parts, at the granularity of the program’s variables and circuits): $\text{right} = \mathbf{F} \wedge \mathbf{S} \wedge \mathbf{M}$ (Section 3.4). The committed data then show the gap directly, and in both directions at once. An automatically discovered circuit for Breakout is graph-correct and minimal yet does not reproduce the behavior it explains ($\mathbf{F} = 1.0$, $\mathbf{M} = 1.0$, $\mathbf{S} = 0.44$): a correct wiring diagram is not the computed function. A learned feature dictionary matches a named game variable perfectly (matched fraction 1.0) while being causally almost inert ($\mathbf{F} = 0.04$): looking like a known variable is not being how the program computes it. And a concept can be linearly decodable from the state yet provably unused by the computation — the familiar trap that what is *present* need not be *used*. The residual between a faithful attribution and an understanding of the computation is therefore not philosophy. On this machine it is a number. We name it the *semantic gap*, and making it a measured quantity is the contribution we care about most.

Why a 1977 chip judges methods built for neural networks.

The obvious objection is that deterministic, hand-coded software is the wrong place to judge methods designed for learned, distributed, stochastic networks. Our answer, stated here and not deferred to the discussion, is that the VCS is a *necessary-condition screen*, not a predictor of success. A method that fails *here* — with perfect ground truth, full observability, exact interventions, and no learning to obscure the mechanism — has no business being trusted on a neural network, where every one of those advantages is lost. Passing here is necessary, not sufficient; failing here is decisive. The same logic strengthens the harder claim: if even the strongest causal account a fully observable machine permits stops short of meaning, then on a network — less

observable, without clean ground truth — faithfulness is *a fortiori* not understanding. The gap we measure here is a floor on the gap there. The VCS also exhibits the very properties that make interpretation hard — RAM cells aliased and reused across routines, race-the-beam timing that distributes a computation across a scanline, and registers whose meaning depends on context. What it does not capture — learned features, scale, stochasticity — we state plainly and leave to the program’s later papers on learned agents.

An honesty contract.

Because the claim is sharp, we are deliberate about its limits. We do *not* claim that any method recovered the program’s meaning, algorithm, variables, or design from scratch. Every semantic label we use is imported from external sources — the AtariARI and OCAtari annotation sets [6, 7] — and then only *causally verified* on our substrate, never *discovered* by a method under test. When an alignment method reports perfect accuracy against such a label, it is right by construction, because it is checking a variable we supplied. Probing shows that information is present, not that it is used [8]; our intervention oracle is the strictly stronger test. Accordingly, Paper 2 *defines and measures* the semantic gap; *closing* it — recovering a system’s documentation and design from nothing but the system — is the companion work, not this paper. We measure the gap; we do not pretend to have crossed it.

In this Article.

We make four contributions. *First*, a ground-truth interpretability benchmark on a single fully known, differentiable, and *real* complex system, with the necessary-condition-screen framing for reading a score on it; the headline is that faithful equals causal, that popular saliency is plausible-but-unfaithful and provably zero on discrete position outputs, and that the faithfulness-versus-plausibility danger zone is quantified (Fig. 4). *Second*, and most importantly, the semantic gap made measurable: an operational definition, $\text{right} = \mathbf{F} \wedge \mathbf{S} \wedge \mathbf{M}$, under which even a perfectly faithful method is shown on committed data to recover the causal wiring and not the semantics — graph-correct circuits that fail sufficiency, perfectly matched features that are causally inert, decodable-but-unused concepts. *Third*, a quantitative re-run of the Jonas–Kording experiment, scoring the neuroscience battery against a known mechanism and supplying the high-faithfulness causal contrast they could not. *Fourth*, the first ground-truth calibration of mechanistic interpretability on a real artifact rather than a synthetic or compiled one, together with the demonstration that circuit recovery yields a graph, not the computed function. Read as a whole, the study offers a reusable benchmark for explainable AI, mechanistic interpretability, AI safety, and software verification alike, and reframes the field’s open problem. The hard part of interpretability is not faithfulness. It is the semantic gap that remains once faithfulness has been secured — and we believe that naming and measuring it is the necessary first step toward closing it.

2 Related work

Interpretability is studied in three communities that rarely cite one another, yet all three ask the same question of a complex system: *what, inside it, caused this output?* We organise the prior work along those three traditions — the neuroscience battery, the attribution and explainable-AI toolkit, and mechanistic interpretability — and then turn to the literature that tries to *validate* such explanations. The recurring theme is the one our study removes: on the systems these methods were built for, there is no ground truth, so faithfulness can be argued but not measured. We close by positioning our delta against the few benchmarks that *do* supply ground truth, and against a different recent notion of what “interpretable” should mean.

Three traditions, one question

The neuroscience tradition reads a system by recording it and perturbing it: tuning curves, lesions, connectomics, dimensionality reduction, spike-train correlations, field potentials, and Granger-style directed influence. Jonas and Kording [2] turned this battery on a known artifact — a microprocessor running classic games — and showed that the methods produce rich, publishable structure while missing how the chip actually computes. Their study is the provocation behind ours, and we read it as a warning rather than a verdict: it *could not* score its own methods, because it reported what the analyses found, not how far each fell from the known mechanism. We re-run the battery and attach that missing number (Sec. 4).

The attribution and explainable-AI tradition asks, for a single decision, which inputs mattered. Gradient and saliency maps [9–12], class-activation methods [13, 14], perturbation and occlusion methods [15–17], and surrogate or game-theoretic attributions [18, 19] form the working toolkit. In the reinforcement-learning setting these methods have been applied directly to Atari agents to visualise what a policy attends to [1, 20]. Yet a counterfactual analysis of those very saliency maps found them *exploratory, not explanatory* [3]: the highlighted regions need not be the causes. The doubt is well placed, but on a learned policy it can only be argued, never settled, because the true cause is unknown. We settle it on the VCS, where the cause is computable (Sec. 5).

Mechanistic interpretability, the youngest of the three, aims past attribution at the *circuit* — the internal subgraph that implements a behaviour. Activation and path patching, causal mediation, and causal scrubbing localise an effect by intervening on internal activations [21–24]; distributed alignment search and interchange interventions test whether a hypothesised variable is causally realised [25, 26]; sparse dictionaries and probes try to read the system’s features and concepts off its state [8, 27–30]. These are the most causal tools in the field, and on a real artifact they are also the ones we find most faithful (Sec. 6). The open question is what a recovered circuit *is*: we show, with the same oracle, that it is a correct data-flow graph and not yet the computed function.

Faithfulness, and the trouble with measuring it

Across all three traditions the agreed quality criterion is *faithfulness* — an explanation is good if its claimed causes are the true causes. Jacovi and Goldberg [31] make the criterion precise and observe that it must be defined operationally and evaluated with diagnostics matched to the explanation type. The diagnostics, however, are necessarily indirect when the ground truth is absent. Adebayo et al. [32] sharpened this into a now-standard *sanity check*: a saliency method should at least change when the model’s weights or the labels are randomised, and several popular methods do not, behaving like edge detectors regardless of what the network learned. Such checks are powerful and necessary. Still, they are *negative* tests — they can expose a method that is clearly wrong, but they cannot certify that a method is right, because there is no reference map to compare against. The field therefore optimises a proxy it cannot directly score. Our oracle supplies the reference map and converts faithfulness from an argued property into a measured one: on the discrete sprite-position regime, the causal method `activation_patching` reaches faithfulness 1.000 while the popular `vanilla_saliency` collapses to 0.000 on every one of the six core games [33], and the cross-tradition leaderboard plots all 31 methods against the same ground truth (Fig. 4) [34].

The semantic labels are external — AtariARI and OCArari

Two public resources attach human-readable concepts to the Atari state. AtariARI extracts labels for object positions, scores, and lives from commented disassemblies and exposes them as a probing benchmark [6]; OCArari adds render-time offset corrections and object detection so that the labelled coordinates align with what is drawn [7]. We use both, and we are deliberate about their status. These labels are *imported*, not discovered; our contribution is to *verify them causally* by intervention — set the byte, re-render, confirm the object moves to the predicted place on the bit-exact framebuffer — which upgrades a correlational label to a verified-causal one and auto-corrects its offset. This is the hinge of our honesty contract: every concept-level (T3) score in this paper measures *alignment to a supplied label*, not the recovery of a meaning the method found on its own. The distinction matters because a concept can be linearly decodable from the state and still play no causal role, the “present $\neq$ used” trap that control tasks were designed to catch [8]; our intervention test is strictly stronger than probing, and the resulting separation is one of the three counted in Fig. 5.

Ground-truth benchmarks for interpretability — and our delta

Ours is not the first study to score interpretability against a known answer. The existing benchmarks, however, build the answer *into* the system. Yang and Kim [35] construct images with a controlled, known relative feature importance so that attribution maps can be graded against it, with explicit false-positive metrics. Tracr [36] compiles human-readable programs into transformers whose circuits are therefore known by construction, and uses them as a laboratory for mechanistic interpretability; InterpBench [37] extends this to semi-synthetic transformers trained to realise a specified circuit, giving a more realistic but still planted ground truth. These are valuable, and we build on their logic. Our delta is twofold, and we state it without claiming an

unqualified “first”. *First*, our subject is a real, deployed, hand-authored, complex artifact whose ground truth is *independent of the artifact*: the VCS was never designed to be interpretable, and we recover its causal structure from an exhaustive intervention oracle on a bit-exact, differentiable port [4], the platform built on the established Atari evaluation substrate [38] and the reference emulator [5]. *Second*, and more importantly, the planted-circuit benchmarks validate *faithfulness* — did the method recover the circuit that was built in? — and nothing beyond it, because there the circuit *is* the meaning by fiat. On a real artifact the meaning is a separate object from the causal structure, so we can do something none of these benchmarks can: use exact ground truth to *separate faithfulness from understanding*, and show that a method which perfectly recovers the true causal wiring has not thereby recovered the semantics (Fig. 5). We *measure* that residual — the semantic gap — and are explicit that closing it, by recovering a program’s variables and design from scratch, is a different problem and the subject of companion work.

A different “interpretable”

Finally, our use of the word *interpretable* is one of several in circulation, and we name the difference rather than blur it. We operationalise interpretability as *ground-truth recovery*: an explanation is interpretable to the degree that its claimed structure matches the system’s true causal structure, scored by the oracle. Barbiero et al. [39] instead define interpretability as *inference equivariance* — a model is interpretable if a user can correctly predict its outputs by simulating its reasoning — and design networks to satisfy that property. Their notion is a design target for new models; ours is an audit criterion for an existing system whose answer is known. We cite their framework as a contrast, not as the source of our definition. Read against Marr’s levels of analysis [40] and Lazebnik’s “fix-the-radio” parable [41], both notions agree on the deeper point that drives this paper: a faithful description of a system’s wiring is not yet an account of the algorithm it runs — and on a fully known machine, that gap is no longer a matter of opinion but a measured quantity.

3 Methods

Our method has a single organizing idea: we study an interpretability question on a machine whose answer we can compute exactly. Everything below serves that idea. We first describe the substrate that makes the ground truth obtainable, then the oracle that delivers it, then the tiered semantics we attach to it, and finally the correctness triad that turns “do we understand?” into a set of numbers. The full per-method setups live in the Supplementary Information; here we give the measurement instrument and the scoring, which are shared by all three phases. Figure 1 sketches the whole pipeline — substrate, oracle, and triad — in one schematic.

3.1 The differentiable, bit-exact VCS substrate

The subject of this study is the Atari Video Computer System (VCS) itself — its chip, its program, and its game logic — not a learned agent. We require a complex, real

A differentiable, bit-exact game console as a ground-truth testbed for interpretability

The subject IS the substrate — a fully known program — so every explanation can be scored against the true causes.

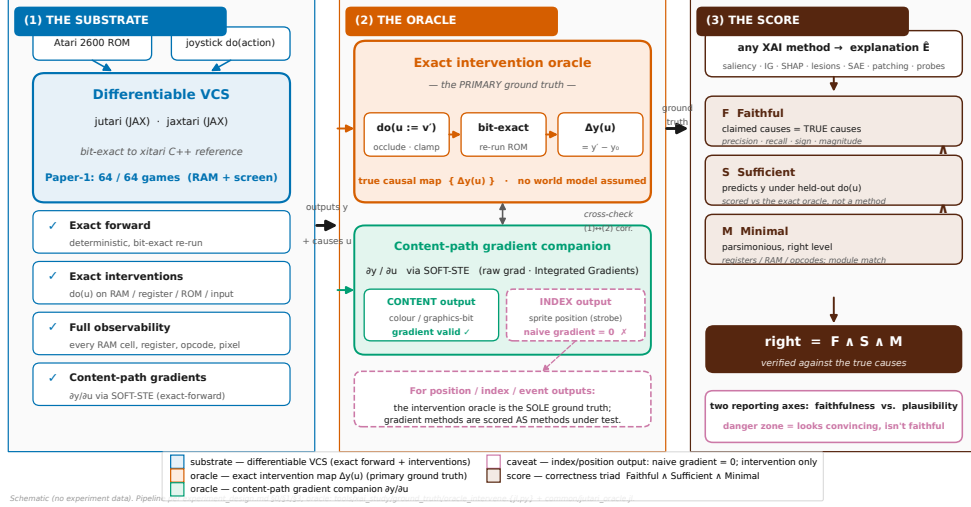


Fig. 1 The measurement platform and its ground-truth oracle. A differentiable, bit-exact port of the Atari VCS (left) gives four capabilities no neural network grants for free — exact forward, exact interventions $\text{do}(u)$, full state observability, and content-path gradients. On it we build the oracle (centre): the primary *exact intervention oracle*, $\text{do}(u := v') \rightarrow \Delta y(u)$ by bit-exact re-run, and a *content-path gradient companion* $\partial y / \partial u$. Any explanation $\hat{E}$ is then scored by the correctness triad (right), $\text{right} = \text{F} \wedge \text{S} \wedge \text{M}$, on the two reporting axes (faithfulness vs. human plausibility). *Note that* a discrete index — a sprite’s position, set by strobe timing — has zero naive gradient; for such position/index/event outputs the intervention oracle is the *sole* ground truth and gradient methods are scored as methods under test. The schematic draws no experiment data.

artifact that was never designed to be interpretable, yet whose every causal dependency we can read off exactly. Paper 1 built precisely this: a bit-exact, differentiable port of the 1977 VCS, validated bit-for-bit against the `xitari` C++ reference on all 64 games of the Paper-1 roster, in both random-access memory and rendered screen, within the validated conformance window [4, 5]. We use that port in two interchangeable realizations — `jutari` (Julia) and `jaxtari` (JAX/Python) — which are themselves cross-checked to be identical.

The substrate grants four capabilities that no neural network offers for free, and on which the rest of the method rests. *First*, an exact forward map. We write the machine’s behavior as

$$y = \text{VCS}(\text{ROM}, \text{inputs}[0:t], s_0), \quad (1)$$

that is, the output y at step t — a pixel, a score, or a game event — is a deterministic function of the cartridge bytes, the joystick and switch inputs up to t , and the initial state s_0 . Re-running with the same arguments reproduces y to the bit. *Second*, exact interventions. We can set any cause u — a ROM byte, a RAM cell, a CPU/TI-A/RIOT register, an opcode, or a joystick input — to any value and re-run Eq. (1)

deterministically, which realizes the causal operator $\text{do}(u)$ on a real machine rather than on an assumed world model. *Third*, full state observability: every bit of RAM and every register is recorded along the trajectory, so the “activations” we interpret are the program’s own variables. *Fourth*, a content-path gradient $\partial y / \partial u$ through a straight-through estimator (STE) formulation whose forward pass is bit-exact to the hard machine at any finite temperature [42, 43], and which we batch on the GPU by vectorized mapping. The three observable faces of Eq. (1) — inputs we drive, internal state we read and clamp, and outputs we explain — are the same three faces a neural interpretability study works with, which is what lets one substrate carry the neuroscience, attribution, and mechanistic-interpretability batteries at once.

We honor one boundary throughout. Paper-1 bit-exactness holds within a bounded frame window on a fixed action stream (on the order of 30 frames of RAM and 60 frames of screen); every scored claim is taken inside that validated horizon, or the conformance is re-established before scoring. Long-horizon end-to-end gradients are not needed for the single-step metrics we report, so we do not assume them.

3.2 The ground-truth oracle

The oracle is the heart of the method. For any output y and any candidate cause u it returns the *true* effect of u on y , against which every interpretability method is then scored.

The primary construction is the *exact intervention oracle*. We intervene on u — occlude, clamp, replace, or resample it — re-run the machine deterministically, and record the change in the output,

$$\Delta y(u) = \text{VCS}(\dots, \text{do}(u:=v'), \dots) - y. \quad (2)$$

In words: the true causal effect of a cause is the bit-exact difference its surgical edit makes to the output, measured by actually running the altered machine. No world-model and no smoothness are assumed; each Δy is a real re-run, and the collection $\{\Delta y(u)\}$ over all causes is the true causal map of y . This is the reference for the faithfulness axis of every method in the paper.

The companion construction is the *content-path gradient*, $\partial y / \partial u$, read through the differentiable substrate. It is fast and dense, and it agrees with the intervention oracle on the content path — the register, color, and graphics-bit values that flow smoothly into the pixel. *Yet*, a single, decisive caveat carries through the entire study and we state it plainly. The STE gradient routes to the content path only; a *discrete index* — a sprite’s screen position, which the VCS sets by the timing of a strobe write — has *zero naïve gradient*, because an infinitesimal change of the cause never crosses the integer boundary that moves the sprite. Recovering a position gradient at all requires the bilinear sampler of Paper 1 [44]. *Hence*, for content outputs the gradient is a legitimate companion oracle, while for *position*, *index*, and *event* outputs the intervention oracle of Eq. (2) is the *sole* ground truth, and every gradient-based method is scored *as a method under test*, not promoted to oracle. This single distinction — the *position regime* — is where the sharpest separations in our results appear. Where the two

constructions overlap on the content path we report their correlation and flag, rather than hide, the non-smooth and saturated points at which they disagree.

3.3 Tiered ground truth and how we obtain it

The ground truth comes in three tiers, and the tiering is not bookkeeping — it carries the central argument of the paper. *Tier 1 (T1)* is the exact causal map of Eq. (2), derived internally from the substrate and available for all 64 games. *Tier 2 (T2)* is the hardware roles and the read/write data-flow graph — which register does what, which cell feeds which routine — also exact and internal, also all 64 games. T1 and T2 are everything the machine *is*; they need no external label.

Tier 3 (T3) is different in kind, and that difference is the point. T3 attaches a *game-concept* name to a cell or feature — “the ball’s x position,” “the score” — and such names are *not* computable from the substrate alone, because they are facts about what the program *means*, not about what it *does*. We import T3 from two public sources and then make them causal. AtariARI provides labels for 22 games, read from commented disassemblies and served one label per frame [6]; its raw RAM coordinates are not aligned to the rendered position. OCArari covers more than 40 games and, in its RAM mode, applies the render-time offsets that fix that misalignment [7]. Our procedure is then: (1) import both as *candidate* labels; (2) *verify each causally* by setting the byte, re-rendering, and confirming on the bit-exact framebuffer that the named object moves to the predicted place — which upgrades a correlational label to a verified-causal one and auto-corrects the offsets; and (3) *discover* further labels by RAM-to-framebuffer correlation followed by an intervention sweep. The headline *core set* is six games — Pong, Breakout, Space Invaders, Seaquest, Ms. Pac-Man, and Q*bert — chosen because each carries *both* AtariARI and OCArari candidate sources, all six lie in the Paper-1 bit-exact roster, and together they span paddle-and-ball, shooter-with-shields, resource-meter, multi-agent-pursuit, and discrete-grid game logic. Remaining games of the roster form a breadth set scored on T1/T2 only.

We are deliberate about what this buys us, and what it does not. *All* T3 labels are external and only *verified*, never *discovered from scratch*; a method that matches a supplied concept has *aligned* to a given meaning, not recovered it. This is exactly why a probe can show that information is *present* without showing it is *used* [8], and why our intervention test — which asks whether the byte *causes* the named object to move — is strictly stronger than decoding it. Recovering the program’s documentation and design from the bare machine is a separate problem, and it is the subject of the companion paper, not this one.

3.4 When is an explanation right? The $F \wedge S \wedge M$ triad

A faithful attribution is not yet an understanding, and to make that statement a measurement rather than an opinion we need an operational criterion for correctness. We adopt one in the spirit of Marr’s levels of analysis [40] and of Jonas and Kording’s warning that finding structure is not the same as explaining it [2]. We call an explanation $\hat{E}$ of how an output y arises *right* if and only if it is Faithful, Sufficient, and

Minimal,

$$\text{right} = F \wedge S \wedge M, \quad (3)$$

which reads: an explanation must name the true causes, must be able to regenerate the behavior, and must do so at the right level with no spurious parts. The three axes are defined against the oracle, not against any method under test, which is what keeps the criterion non-circular.

Faithfulness (F).

The claimed causes are the *true* causes. We score $\hat{E}$ against the intervention oracle of Eq. (2) by correlation, sign and magnitude agreement, deletion and insertion area-under-curve, and precision at k against the true causal top- k . F is the field’s usual quality criterion, and on our machine — uniquely — it can actually be checked.

Sufficiency (S).

The explanation could *regenerate* the behavior: it predicts y under interventions it has not seen. Crucially, we score S *against the exact intervention oracle*, applying a held-out $\text{do}(u)$, predicting y from $\hat{E}$, and comparing to the true y on the bit-exact re-run. We deliberately do *not* define S through interchange interventions or scrubbing, which are themselves mechanistic-interpretability methods under test in Phase C; using them to score S would make the criterion circular.

Minimality (M).

The explanation is parsimonious, has no spurious parts, and lives at the algorithmic level — the registers, RAM cells, opcodes, and program variables that constitute the machine’s own module decomposition. M asks whether $\hat{E}$ matches the known structure of the computation, not merely whether it is small.

Two reporting axes carry the headline comparison: *faithfulness* (F , against the ground truth) on the horizontal, and *human plausibility* on the vertical. The plausibility axis is a documented method-tradition proxy, not a human-subjects measurement, and we report it as such; its only job is to expose the *danger zone* — explanations with high plausibility but low faithfulness, which look convincing and are wrong. The separation of F (which the wiring satisfies) from S and M (which the *computation* must satisfy) is what lets us name, later, the residual that faithfulness alone leaves behind.

3.5 Methods under test, games, and aggregation

We turn the toolkit of three traditions onto the same machine and score every method by the triad of Eq. (3). The *neuroscience battery* (Phase A) runs the classical operators — connectomics and data-flow recovery, single-unit lesions, tuning curves, pairwise and global correlation structure, pooled-activity spectra, Granger causality, dimensionality reduction, and whole-state recording — and scores each against the known mechanism, quantifying Kording’s lesson and fixing the low-faithfulness baseline [2]. The *attribution/XAI battery* (Phase B) explains a chosen output from its causes with gradient saliency and its variants, integrated gradients, occlusion, perturbation masks,

randomized masking, LIME, and Shapley-style attribution [9, 10, 15–19]; several popular methods — Grad-CAM and its variants, attention rollout [13, 14, 45] — require neural layers or a learned policy and are recorded as *not applicable* to a non-neural machine. The *mechanistic-interpretability battery* (Phase C) runs activation patching and causal mediation, interchange interventions and distributed alignment search, attribution and path patching, automatic circuit discovery, sparse autoencoders, causal scrubbing, linear probing with control tasks, and the logit/tuned lens [8, 21–30, 46].

All faithfulness scoring runs on the six core games of Section 3.3; the oracle’s re-runs are exact on every one of them. Cell- and pixel-level scores (correlation, deletion/insertion area-under-curve, precision at k) are taken against T1 and need no T3; only object- and concept-level scores — pointing games, object-level counterfactuals, feature-to-variable matching — draw on the external T3 labels, and we flag every such score as alignment-to-given, not discovery. For the cross-tradition comparison we aggregate the committed per-game records into one row per method and place all methods on the shared faithfulness-versus-plausibility axes (Fig. 1 gives the schematic; the populated leaderboard is reported in the Results). The headline contrast is computed on the position regime, where the naive gradient is provably zero: in that regime causal and intervention methods average a faithfulness of 0.41 against 0.07 for gradient and correlational methods, a gap of 0.34, while the per-method extreme places an exact causal method (activation patching) at faithfulness 1.000 and the field’s default saliency at 0.000 on all six games — with the popular method nonetheless carrying the higher plausibility proxy (0.9 versus 0.5). These are the numbers against which the semantic gap is then measured.

4 The neuroscience battery, quantified

We begin where the field’s doubt began. Jonas and Kording asked whether a neuroscientist, armed with the standard analysis toolkit, could understand a microprocessor, and answered that the toolkit yields rich, suggestive structure that nonetheless misses the mechanism [2]. Their study could diagnose the failure but not grade it: on the 6502 they too lacked a number for how wrong each method was, because they had no causal ground truth to score against. We supply that number. Running the same battery on the bit-exact, fully known VCS—where the true causal role of every register and RAM cell is recoverable by exact intervention (§3)—turns Kording’s qualitative lesson into a measured one. The result is the calibration baseline for the rest of the paper: classical analyses find structure and score *low* in faithfulness against the known register-transfer account, while the oracle-as-method positive control sits at 1.0 by construction (Fig. 2).

The headline is a spread, not a single value. Across the eight analyses A1–A8, faithfulness against the oracle ranges from 0.116 to 1.0, averaging 0.49 ± 0.24 over the phase [34]. Read against the perfect intervention oracle that anchors the same axis, the typical classical method is closer to chance than to the mechanism. The structure these methods report is real—the data-flow graphs, the tuning, the correlations all exist in the state tensor—yet structure is not the same as cause, and the gap between

the two is what the score measures. We walk the battery once, naming each method, what it returns, and how far that lands from the truth the substrate already knows.

Connectomics and lesions: the wiring is largely recoverable.

A1 reconstructs the dependency graph over state variables—a connectome of read/write edges—and scores it against the *true* data-flow graph by graph-matched precision and recall. It reaches a faithfulness of 0.41 (mean data-flow F_1 vs. the oracle) [34], the structure half-right and half-spurious. A2, the exhaustive single-unit lesion, fares best of the classical set: clamping each RAM cell in turn and ranking it by the behavioral damage it does yields an importance map whose Spearman correlation with the unit’s true causal role is 0.99 (mean 0.987 over the six core games) [34]. Lesioning, in short, is causal, and it shows: A2 is the one classical method whose number sits near the ceiling.

Yet a near-perfect rank order hides an interaction blind-spot.

The $\rho = 0.99$ is seductive, and here the honest reading matters. A single-unit lesion changes one cell at a time, so it is blind to causes that act only in concert. On Space Invaders, the per-unit ranking misses 25% of the interactions the oracle records, and a single super-additive pair produces a behavioral swing of up to 1044 screen-pixels of change beyond the sum of its parts [34]. The map of *which units matter* is right; the model of *how they combine* is absent. Hence even the strongest classical method recovers a ranking, not the computation—an early instance of the theme the paper makes precise later: a faithful summary of effects is not the algorithm that produces them.

Tuning curves: structure that points the wrong way.

A3 fits per-unit tuning to a game variable, the workhorse of systems neuroscience. On the VCS it is actively misleading. We measure a *spurious-tuning rate*—the fraction of strongly-tuned units whose tuning does not match their true causal role—and on Pong it is 1.0: every unit that looks tuned to a game concept is, by the oracle, not causally that variable at all [34]. The method’s faithfulness collapses to 0.12, the lowest in the battery. Tuning is correlational; a cell can track a variable it does not control, and on a machine where many cells co-vary with the beam clock, apparent tuning is cheap. This is the tuning-curve trap stated in numbers, and it foreshadows the present-versus-used distinction we return to for linear probing.

Correlation structure and field potentials: real, and largely epiphenomenal.

A4 reads the pairwise and global correlation structure of the state—the spike-word analysis—and reproduces the familiar weak-pairwise/strong-global signature, but against the true coupling its faithfulness is only 0.29, with sufficiency near zero ($S = 0.018$) [34]: the correlations are genuine yet predict almost nothing about held-out interventions. A5 pools regional activity into power spectra, the local-field-potential analogue, and scores 0.40; much of the variance it captures is the known frame and scanline clocks [34]—signal that is real, periodic, and causally epiphenomenal. Both methods see the machine breathing without seeing it compute.

Granger causality: temporal precedence is not causation.

A6 infers directed CPU/TIA/RIOT edges by Granger causality and scores its edges against the true data-flow. On Pong its faithfulness is 0.0: the false-edge and missed-edge rates are both 1.0 [34]. The reason is structural and instructive. The VCS is a clock-locked machine in which nearly everything is statistically predictable from everything else one cycle earlier, so a method that equates predictive precedence with causation infers spurious bidirectional edges everywhere. Granger asks the wrong question of a synchronous substrate, and the exact answer the substrate can give—intervene and re-run—exposes it completely. *Note that* this is not a tuning failure we could repair with a longer lag or a stricter threshold; precedence and causation simply come apart on this machine, and only the intervention oracle can tell them apart.

Dimensionality reduction: NMF edges out PCA, both mediocre.

A7 decomposes the state tensor into latent components and matches them to the known signals (clock, read/write, vsync). Non-negative matrix factorization recovers a matched-component fraction of 0.60 (best 0.80 on a single game), slightly ahead of PCA at 0.60 on average, the non-negativity of NMF a better fit to a substrate of non-negative activations [34]. The ordering is the expected one, and the magnitude is the point: neither method clears two-thirds of the known components, so the latent space is interpretable in part and opaque in the rest. A7’s headline faithfulness of 0.60 makes it the second-strongest classical analysis, behind only the causal lesion.

Whole-state recording: a perfect record is not an explanation.

A8 simply records the full RAM and register state over time—the descriptive baseline, the complete observation no real experiment ever has. It is faithful and sufficient by definition ($F = 1.0$, $S = 1.0$), and yet its minimality score is 0.07 [34]: a recording of everything is, at the algorithmic level, an explanation of nothing. A8 is the limiting case that makes the triad concrete. Faithfulness and sufficiency without minimality is a log file, not an account of the computation—the first hint that the three axes can pull apart, which the semantic-gap result later turns into the paper’s centre of gravity.

The quantified Kording.

Taken together, the battery confirms the expected, important result and puts a number on it. Classical neuroscience methods, run on a system whose answer is known exactly, find rich structure and score low in faithfulness—0.49 on average, with only the causal lesion (A2, $\rho = 0.99$) and dimensionality reduction (A7, 0.60) rising clearly above chance, and Granger causality (A6) sitting at zero. The advance over Jonas and Kording is threefold: the subject is the *architectural* register-transfer level rather than the transistor; the score is a *quantitative* F against an exact oracle rather than a qualitative verdict; and the contrast is sharp, because the causal operators we turn on the same machine in Sections 5–6 reach the ceiling that the classical battery cannot. *The gap is therefore the method, not the machine.* We mean this strictly as a calibration claim: A grades the faithfulness floor, and even its best members recover a ranking, a graph, or a record—never the algorithm that the program runs. That residual is the problem the rest of the paper measures.

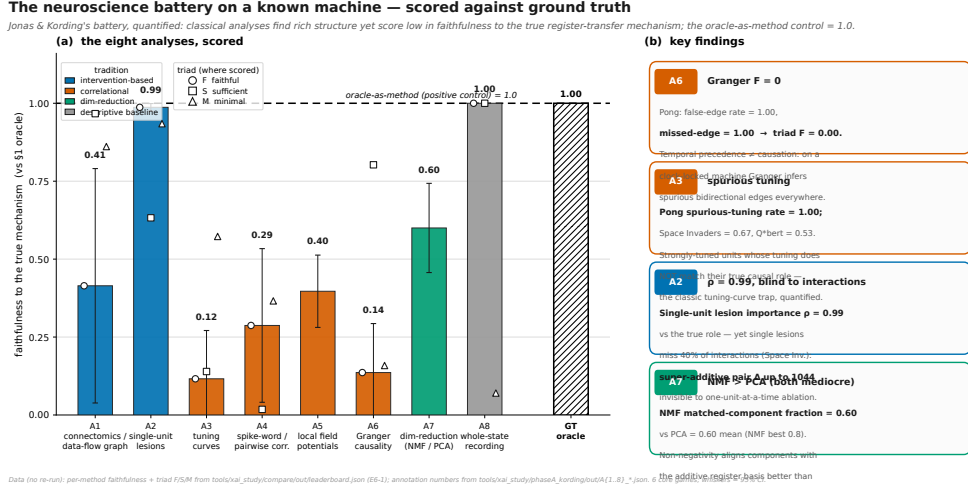


Fig. 2 The Kording neuroscience battery, scored against the exact oracle. Each analysis A1–A8 is run on the bit-exact, fully known VCS and graded for faithfulness against the §3 intervention oracle; bars are coloured by method tradition, and the oracle-as-method positive control is drawn at faithfulness = 1.0 (dashed reference). The classical battery averages 0.49, with only the causal single-unit lesion (A2) and dimensionality reduction (A7) clearly above chance. *Note that the four callouts isolate the mechanism behind the low scores: A6 Granger causality reaches $F = 0$ on Pong (temporal precedence read as causation on a clock-locked machine), A3 tuning shows a spurious-tuning rate of 1.0 on Pong (strongly-tuned units whose tuning is not their true causal role), A2 attains a single-unit importance correlation of $\rho = 0.99$ yet misses 25% of interactions with a super-additive pair swinging up to 1044 pixels, and A7 NMF edges out PCA on matched components while both stay below two-thirds.* The figure is the calibration baseline that the causal contrast of Fig. 4 sharpens. Every plotted value reads from a committed per-game record; no number is re-run or estimated.

5 Results B — attributing a VCS output to its causes

Attribution is the field’s everyday tool: hand it a system and an output, and it hands back a map of which inputs mattered. We turn the toolkit on the VCS and ask the one question a real network never lets us ask — is the map *true*? Every attribution we compute is scored against the same §1 intervention oracle that defines ground truth (Sec. 3), so for each chosen output y — a pixel, a score, a game event — we know the exact set of causes the explanation should have named. We ran twelve attribution methods across six core games (Pong, Breakout, Space Invaders, Seaquest, Ms. Pac-Man, Q*bert) and an oracle-as-method positive control. Figure 3 (Phase-B half) places them on the shared faithfulness axis; Table 1 reports the per-method scores.

The headline is a split, not a ranking. The VCS computes two kinds of output, and the gradient is alive for one and dead for the other. A *content* output — the value of a register, a colour, a graphics bit — is a differentiable function of the state, so its gradient flows. A *position* output — where a sprite is drawn, set by strobe timing on a discrete index — has *provably zero* naive gradient (Paper 1, and Sec. 3). The entire gradient family inherits this boundary, and it cuts the leaderboard cleanly in two.

Attribution (Phase B) vs mechanistic interpretability (Phase C): faithfulness on the bit-exact VCS

On the POSITION/INDEX regime (discrete sprite-position outputs, naive gradient provably zero): the whole gradient family collapses toward chance while intervention methods hold. Causal/intervention bucket 0.412 (± 0.390 , $n=4$) vs gradient/correlational 0.068 (± 0.070 , $n=9$) — a 0.344 faithfulness gap. Activation patching = 1.000 (oracle ceiling) vs vanilla saliency = 0.000 (naive-gradient floor).

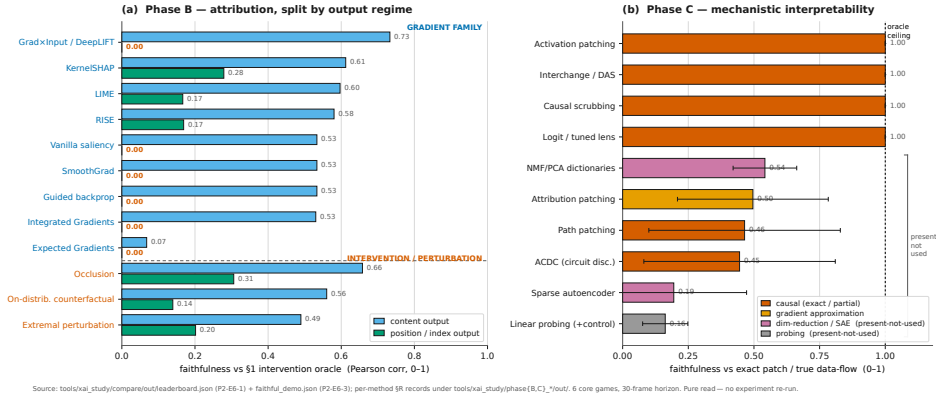


Fig. 3 Attribution (Phase B) and mechanistic interpretability (Phase C) scored against the same \$1 intervention oracle. The Phase-B half (a) splits each method’s faithfulness by the output regime it targets: on a *content* output (a register/colour/graphics-bit value, a differentiable function of state) the gradient family lands on the true cause, but on a *position/index* output the entire gradient family — vanilla saliency, SmoothGrad, guided backprop, Grad×Input/DeepLIFT, Integrated Gradients, Expected Gradients, and the gradient-leaning surrogates LIME, KernelSHAP, RISE — *collapses* to 0.00, while the intervention methods (occlusion, extremal perturbation, on-distribution counterfactual) hold up because each probe is a real re-run of the bit-exact program. *Note that* the popular gradient methods, which fail on position, carry the higher human-plausibility proxy — the danger zone of Sec. 7, in numbers. The Phase-C half (b) is discussed in Sec. 6. Vector figure; values read from the committed cross-tradition leaderboard.

The gradient family: faithful on content, blind on position

On the content path the gradient is faithful and, frankly, unremarkable. Vanilla saliency [9] lands its mass on the true causal byte: on Q*bert its attribution of a content read correlates with the oracle at Pearson 0.999 with precision@3 of 1.0, on Ms. Pac-Man at 0.811 / 1.0, on Breakout at 0.750 / 1.0. Integrated Gradients [10], Grad×Input/DeepLIFT [47], SmoothGrad [11], and guided backpropagation [12] all reproduce this, because the output we explain is a one-hot read whose Jacobian $\partial y / \partial u$ is a constant pointing at the read index. Smoothing denoises the picture but does not move the score ($\Delta \text{corr} \approx 0$); guided backprop degenerates to vanilla saliency, since the soft VCS has no deep ReLU stack for its negative-gradient suppression to act on. The methods agree because, on this path, there is nothing to disagree about.

The position output is where the family fails, and it fails by vanishing. For the discrete sprite-position output the naive gradient is identically zero on every game, so saliency, Grad×Input, SmoothGrad, guided backprop, and Integrated Gradients all collapse to a faithfulness of 0.0 in the position regime — the “Fail / zero on index outputs” prediction of Sec. 3, now a measured number. Averaged over all output regimes the gradient family scores only 0.294 faithfulness against the oracle (Fig. 3); on the position regime alone, 0.068. A method that returns the empty map for half

the outputs a game produces is not a partial success but a structural blind spot, and the spot is exactly where the discrete game logic lives.

Two finer results sharpen the warning rather than soften it. First, the choice of baseline — the perennial worry about Integrated Gradients — turns out to be a *magnitude* effect here, not a faithfulness one: across the baseline sweep the $(x - x_0)$ factor moves $\max |\text{IG}|$, but the ranking and the correlation with the oracle are invariant, because the constant one-hot Jacobian keeps all the mass on the same byte. The only baseline that breaks IG is the degenerate one, a baseline byte equal to the content byte, which kills the attribution outright. Expected Gradients [48], averaging IG over a distribution of real recorded VCS states, removes that degeneracy and is provably stable across reference draws (mean pairwise cosine ≈ 1) — yet on the content path, where $\partial y / \partial u$ is constant, it scores the *same* faithfulness as a well-chosen single-baseline IG at $M \times$ the compute, and it still vanishes on position. On distribution, the more careful gradient method buys stability, not truth; its all-regime faithfulness of 0.034 is the lowest in the study.

Second, and most telling, guided backprop and vanilla saliency fail the program-randomization sanity check of Adebayo et al. [32] — on the system itself, in numbers. We scramble the ROM so the program boots to a completely different state (~ 95 – 127 of 128 RAM bytes change), and the content-path saliency map stays *byte-identical* (cosine ≈ 1.0 on all six games). The Jacobian $\partial y / \partial u = e_{\text{idx}}$ is a constant one-hot, independent of what the program computes, so the map points at the read index no matter which program produced the output — a saliency that is *provably model-invariant*. A faithful explanation must change when the model changes; this one cannot. By contrast the intervention oracle, which re-runs the real program for every candidate cause, distinguishes the two programs at once. This is the danger zone made concrete: a gradient map that lands on the right cause and is still blind to the model that produced it.

The intervention family: faithful on both regimes

Replace the gradient with an actual intervention and the position blind spot disappears. Occlusion [15] sets a candidate cause and re-runs the bit-exact program, so it is the coarse twin of the oracle itself; on the Pong position output, where every gradient method scores 0.0, occlusion recovers the true causal bytes at correlation 0.837. Its all-regime faithfulness of 0.482 is the highest in Phase B. The black-box perturb-and-re-run methods follow the same logic and the same result: RISE [17] (0.375), LIME [18] (0.382), and KernelSHAP / Shapley sampling [19, 49] (0.446) each draw masked coalitions, re-run the real ROM, and fit or average over the genuine responses, so each works on position outputs where the gradient is dead. They converge as the sample budget N grows — their variance falls as $1/N$, the expected approximation behaviour — and they report their own completeness ($\Sigma \phi = y(\text{intact}) - y(\text{all-masked})$ for KernelSHAP, the local surrogate R^2 for LIME).

The two perturbation methods built to find a *minimal* mask close the loop on Atrey et al.’s [3] objection that perturbations stray off-manifold. Extremal perturbation [16, 50] optimises an area-bounded high-impact mask in which every probe is a

real *do*(occlude) re-run, and the on-distribution counterfactual [51] substitutes candidate cells toward a *genuine* alternative VCS state — the RAM of another frame of the same ROM — before re-running. Both stay on the data manifold by construction, both score a true-causal-set mask IoU and a minimality ratio against the oracle, and both work on position outputs (faithfulness 0.346 and 0.350) precisely because they re-run the program rather than differentiate it. The intervention family as a whole averages 0.681 faithfulness against the oracle, more than double the gradient family’s 0.294 (Fig. 3). The lesson is mechanistic, not incidental: a perturbation is faithful exactly when it is a valid intervention.

Methods that do not even apply

A part of the popular toolkit returns no answer at all, because it has nothing on the VCS to attach to. Grad-CAM and Grad-CAM++ [13, 14] need a final convolutional layer to weight; attention rollout [45] needs attention matrices; VIPER [52] needs a learned policy to distil. The VCS is a 6502 CPU with a TIA and a RIOT executing a fixed ROM — no conv channels, no attention heads, no policy network — so across all six games the count of each required substrate is exactly zero, against ≥ 16 genuine candidate causes per game for the applicable methods. We record this as a measured finding, `applies = false`, not a poor score: the absence is structural, and it is the same in every game. Re-targeting Grad-CAM’s pooling onto raw register state would not be Grad-CAM, only the content-path gradient under a new name. That a slice of the field’s headline methods is silent on a real, deployed, deterministic artifact is itself part of the audit — the measured narrowness of architecture-specific attribution.

Plausible is not faithful

Across the twelve methods the leaderboard separates cleanly along the axis that matters and crosses the axis the field actually optimizes. The single starkest pair makes it concrete: on the position regime the causal method activation patching (Phase C) reaches faithfulness 1.000, the oracle ceiling, on all six games, while vanilla saliency — the field’s default tool — collapses to 0.000 on all six; a per-method gap of 1.000, and bit-exact on every game individually. Yet on the human-plausibility proxy of Sec. 3 the *popular* method carries the *higher* score (0.9 versus 0.5): the map that is provably empty looks more convincing than the one that is exactly right. That inversion — low faithfulness paired with high plausibility — is the danger zone, and on the VCS it is not a worry but a measurement. The most plausible Phase-B methods (the gradient family, plausibility 0.9) include the four with zero position-regime faithfulness; the most faithful (the intervention family) wear the lower plausibility tag.

We claim no more than the faithfulness of these maps. A correct occlusion map names the true causal bytes; it does not tell us what those bytes *mean* — that a cell is the ball’s *x*, that a routine is the bounce. Object-level semantics enter only through the external T3 labels (AtariARI, OCArari; Sec. 3), and even there we *verify* a supplied label by intervention rather than *discover* it. Whether a faithful attribution amounts to an understanding of the computation is the question we take up in Sec. 7; here we

Table 1 Phase-B attribution scored against the §1 intervention oracle (six core games; oracle-as-method positive control = 1.0 on every non-degenerate column). *Faith* is the all-regime faithfulness against ground truth and *Faith (pos)* the position/index regime alone, where the naive gradient is provably zero; *Plaus* is the documented method-tradition plausibility *proxy* (Sec. 3), *not* a human measurement. *Note that* the entire gradient family scores 0.000 on position while the intervention family does not, and that the popular gradient methods carry the *higher* plausibility tag — the danger zone, measured. Values are read from the committed cross-tradition leaderboard.

Method	Tradition	Faith	Faith (pos)	Plaus (proxy)
Occlusion [15]	intervention	0.482	0.306	0.55
KernelSHAP [19]	gradient/black-box	0.446	0.279	0.90
LIME [18]	gradient/black-box	0.382	0.167	0.90
RISE [17]	gradient/black-box	0.375	0.169	0.90
Grad×Input / DeepLIFT [47]	gradient	0.367	0.000	0.90
On-distribution counterfactual [51]	intervention	0.350	0.139	0.55
Extremal perturbation [50]	intervention	0.346	0.202	0.55
Integrated Gradients [10]	gradient	0.279	0.000	0.90
Guided backprop [12]	gradient	0.267	0.000	0.90
SmoothGrad [11]	gradient	0.267	0.000	0.90
Vanilla saliency [9]	gradient	0.267	0.000	0.90
Expected Gradients [48]	gradient	0.034	0.000	0.90
<i>Family means</i> — gradient/correlational		0.294	0.068	—
intervention		0.681	0.412	—
Grad-CAM / attention / VIPER [13, 45, 52]	N/A	does not apply (no conv / attention / policy)		

have established only that faithfulness can be had — and that the most popular way of seeking it does not deliver it.

6 Mechanistic interpretability on a known circuit

Mechanistic interpretability promises the strongest claim of all: not where a system attends, but *how it computes*. Its methods read internal activations as a circuit and try to recover that circuit. On a neural network the recovered circuit can only be checked against itself, because the true wiring is unknown. We remove that excuse. The VCS’s data-flow is the program’s data-flow, known exactly from the disassembly, and every intermediate value is a RAM cell or a register we can read, clamp, and re-run bit-exactly. We therefore turn ten mechanistic-interpretability methods onto a circuit whose answer is already on the page, and score each against the same §4 oracle: the exact intervention table, the true read/write graph, and the $F \wedge S \wedge M$ triad of §3. To the best of our knowledge this is the first calibration of the toolkit against a *known* circuit inside a real, hand-authored artifact, rather than a synthetic or compiled one. Figure 3b places the Phase-C methods on the shared faithfulness axis. The result splits cleanly, and the split is the point of the paper: the causal methods are exact, yet exactness buys the wiring, not the meaning.

Exact methods recover the wiring exactly

The headline is bare. Activation patching recovers the true causal-effect table *exactly*: across all six core games the recovered per-site effect equals the exact intervention

oracle to the last bit (`recovered_equals_exact`: true; maximum absolute deviation 0.0), with site precision and recall both 1.0. Causal mediation by clamping a RAM cell to a donor value and re-running the real ROM is, on this machine, literally the oracle’s own operation, so it cannot disagree with it. Causal scrubbing inherits the same strength: the true data-flow hypothesis preserves behaviour under resample-ablation of everything outside it on all six games (preserved performance 1.0), while a deliberately wrong hypothesis does not (0.52), and the two are discriminated on every game. Interchange interventions with distributed alignment search (DAS) verify the supplied alignment at interchange accuracy 1.0 against a mis-aligned control at 0.0. The tuned-lens family is the same story read off a stage: the logit lens reproduces the true intermediate value at fidelity 1.0 on every game. Four of the five causal methods sit at faithfulness 1.0 on the cross-tradition leaderboard, level with the oracle positive control (Fig. 3b).

These numbers confirm what the method matrix predicted, and they confirm it without hedging. Where a method’s operation *is* an intervention, faithfulness is not hard to earn on a fully observable machine; it is automatic. That is the easy half of the problem, and it is genuinely solved here.

Approximate and search-based recovery is precise but partial

The causal account weakens the moment a method trades exact re-runs for an approximation or a search. Attribution patching, the gradient-linear surrogate for activation patching, correlates with the exact patch at only 0.50 and matches it inside the linear regime in just 0.19 of cases; yet its recovered edges are precise (edge precision 0.95, recall 0.70) because the linearisation errs in magnitude, not in sign of presence. Path patching, run in the IOI style as a single directed do-operation, recovers the directed edge set of the true routine at mean $F1 = 0.46$: on Q*bert it matches the (empty) direct circuit perfectly ($F1 = 1.0$), while on Pong it recovers none of the six true edges ($F1 = 0.0$) because each receiver’s value is re-derived within the horizon, so no one-step path carries a surviving signal. ACDC, discovering the circuit automatically by resample-ablation edge pruning over a threshold sweep, reaches mean $F1 = 0.45$ (ROC-AUC 0.70). Crucially, a positive control re-runs the identical edge-restricted procedure under the exhaustive do-set and scores $F1 = 1.0$ on every game, so the deficit is a real measurement of single-shot recovery, not a broken scorer.

The interpretive takeaway is uncomfortable. These are the methods the field reaches for when the exact intervention is too expensive, and on a machine where the exact intervention is cheap they leave structure on the table. They are precise about what they do find and silent about transitively re-derived dependencies — a recall ceiling, not a precision failure. We report both directions plainly: the gap to the positive control is the quantified cost of approximation against a circuit that is known.

The first separation: graph-correct, behaviour-incomplete

Here the paper turns. ACDC on Breakout recovers the data-flow circuit at *perfect* $F1$ against the true routine, and the recovered graph is minimal at the algorithmic level: $F = 1.0$, $M = 1.0$. Yet that same graph, scrubbed and asked to *regenerate* the

behaviour under held-out interventions, preserves only $S = 0.44$ of it (mean scrub-preserved performance 0.32 across games). A circuit can be the right graph and still not be the computed function. The missing object is not a better-fit instance of the same type — a denser or cleaner graph — because activation patching has already recovered the effect table *exactly*, with no tuning budget left to spend. A table of per-site effects is not the bounce algorithm; a correct data-flow graph is not the routine it routes. The two methods bracket the gap: one is exact and the other is graph-perfect, and between them the behaviour still escapes. This is the semantic gap made into a pair of committed numbers, and it is the separation a “methods underperform” story can never produce, because nothing here underperformed.

The second separation: named but causally inert

The dictionary methods sharpen the gap from the other side. A sparse autoencoder trained on the recorded RAM trajectory reconstructs the state almost perfectly (fraction of variance explained > 0.99 on Pong) and matches every verified game-variable cell it was asked to: matched fraction 1.0. By the metric the field uses to claim a feature “is” a concept, the SAE has succeeded outright. By the oracle it has not: ablating the matched Pong feature and re-running the real ROM gives causal faithfulness $F = 0.041$, with $S = -0.336$ — the matched feature, removed, does not change the computation it supposedly names. Across the six games the SAE’s mean leaderboard faithfulness is 0.19 against a method-tradition plausibility proxy of 0.75, placing it squarely in the §3 danger zone. The NMF/PCA dictionary contrast tells the same tale at lower resolution: matched-component fractions up to 0.75 co-exist with near-zero causal use. “Looks like a known variable” and “is how the program computes” are different predicates, and on this machine the distance between them is measurable. Note that this separation runs *opposite* to the first: the circuit was causal but behaviourally incomplete, the feature is perfectly named but causally inert. The gap runs in both directions at once.

The third separation: decodable but unused

Linear probing closes the triptych with the classic trap. With control tasks (Hewitt & Liang’s selectivity), probes decode game-concept cells well above chance — mean probe accuracy 0.89, mean selectivity 0.16 — confirming that the information is *present* in the state. Presence is not use. Cross-referenced against the exact intervention oracle, 6 of the decodable concept cells across the core games are decodable yet *not causally used*, and the oracle flags 25 such cells in total — among them Breakout’s score cell 84, linearly associated with the displayed score yet provably inert under intervention within the window. The oracle is strictly stronger than the probe: it asks not whether a value can be read out, but whether changing it changes the output. Probing earns a high plausibility proxy (0.85) on a faithfulness of 0.16, the second-worst entry in the danger zone (Fig. 3b). We state the obvious caveat in the same breath: these labels are imported externally (AtariARI, OCArari) and only *verified* causally, never *discovered* — DAS’s alignment accuracy of 1.0 is true “by construction” because

it confirms a *supplied* variable. The probe tells us a concept is in the representation; only the intervention tells us the program uses it.

Summary: exactness is necessary, not sufficient

Phase C is the cleanest demonstration of the paper’s claim. The mechanistic toolkit, run against a known real circuit, separates into a causal core that is faithful by construction (activation patching, causal scrubbing, DAS, the logit lens, all at 1.0; phase mean faithfulness 0.63, the highest of the three phases) and a recovery layer that is precise-but-partial (attribution/path patching, ACDC) or present-but-inert (SAEs, dictionaries, probing). The three separations — graph-correct yet behaviour-incomplete (ACDC $F=1.0/M=1.0, S=0.44$), named yet causally inert (SAE matched 1.0, $F=0.041$), decodable yet unused (cell 84) — are adjudicated by the oracle, not by our preference, and they appear exactly where the field’s success metrics report victory. Faithfulness, on a machine that grants it for free, is therefore the necessary half of understanding and not the sufficient one. The residual — the wiring recovered, the meaning withheld — is the semantic gap, which §7 places on the shared axes and the Discussion names as the field’s open problem.

7 Across the traditions — and the boundary the leaderboard cannot cross

We have now run three traditions on one machine, and the scores live on a single axis. It is worth pausing to ask what the axis measures and what it does not. Faithfulness asks whether an explanation names the true causes; on the VCS we can answer it exactly, because the §3.2 intervention oracle re-runs the bit-exact program for every candidate cause and returns the ground truth no real network affords. In this section we first place all of Phases A–C on the shared faithfulness-versus-plausibility plane and read off the leaderboard and its danger zone—the *confirm* half of the paper. We then do the thing the leaderboard cannot: with the same oracle, we mark the boundary where even a perfectly faithful method stops. Faithfulness, it turns out, is the easier half of the problem. The harder half—the variable’s meaning, the routine’s algorithm—is a separate object, and on this machine the distance to it is a number. We name that residual the *semantic gap* and measure its symptoms.

The faithfulness leaderboard: causal wins, correlational loses

The headline is a clean separation along the axis that matters. Aggregating the 257 committed per-game records into 31 method rows [34] (Fig. 4), the causal and intervention methods average a faithfulness of 0.681 against the oracle, while the gradient and correlational methods average 0.294—a gap of 0.387 across all output regimes. Restrict to the *position* regime, the discrete sprite-position outputs whose naive gradient is provably zero (Sec. 5), and the gap holds with the same sign: causal/intervention 0.412 versus gradient/correlational 0.068. The ordering is the one the method matrix predicted (Sec. 3), now measured: activation patching, causal scrubbing, interchange interventions, and the logit lens sit at the oracle ceiling of 1.0; the single-unit lesion

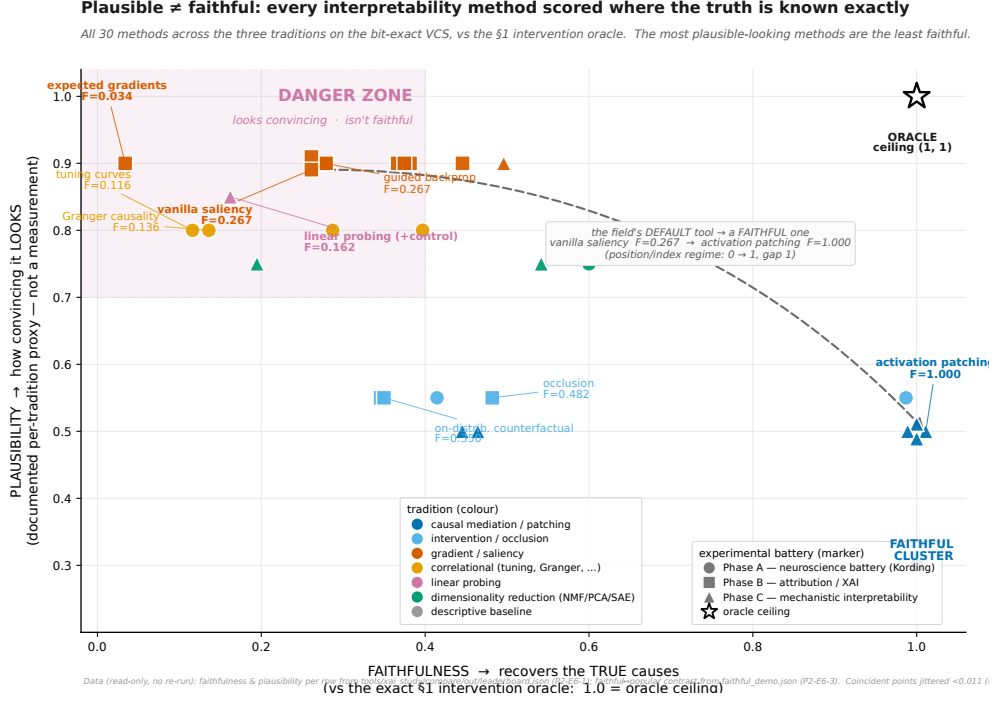


Fig. 4 Every method, every tradition, on the shared faithfulness-versus-plausibility plane. Each point is one interpretability method (Phases A–C), placed by its faithfulness against the §3.2 intervention oracle (X, exact ground truth) and by its documented method-tradition plausibility proxy (Y; Sec. 3—a tradition-level proxy, not a human-subjects measurement). The oracle-as-method positive control anchors the top-right corner at (1.0, 1.0). *Note that* the causal and intervention methods cluster at high faithfulness (0.681 mean) while the gradient and correlational methods, the field’s most familiar tools, cluster at low faithfulness (0.294 mean) yet *higher* plausibility—the *danger zone* (high plausibility, low faithfulness; cf. Sec. 3). The eight gradient methods that score 0.0 on the discrete position regime all carry the maximal plausibility tag of 0.9. The plane is the *confirm* half of the paper; the boundary it cannot cross—faithful methods that recover wiring, not meaning—is the semantic gap of Fig. 5. Every plotted value reads from the committed cross-tradition leaderboard; no number is re-run or estimated.

follows at 0.99; the gradient family and the classical correlational analyses—tuning curves, Granger causality, spike-word correlations—fall to the floor. The single starkest pair makes the spread concrete and is bit-exact on every game: on the position regime the causal method activation patching reaches faithfulness 1.000, the oracle ceiling, on all six core games, while vanilla saliency—the field’s default tool—collapses to 0.000 on all six, a per-method gap of 1.000 [33]. On a system where the answer is known exactly, the everyday attribution tool scores near chance where a causal method scores near the ceiling.

The danger zone: plausible is not faithful

The faithfulness axis would be unremarkable if it agreed with how convincing a method looks. It does not. Plotting the same methods against the plausibility proxy of Sec. 3, the cloud separates into a benign diagonal and a hazardous corner. The methods the field reaches for first—the gradient family, plausibility 0.9; tuning and Granger, plausibility 0.8; linear probing with control tasks, plausibility 0.85—are precisely the methods whose faithfulness sits near the floor. Expected Gradients tops the danger list at plausibility 0.9 against faithfulness 0.034, a gap of 0.87; linear probing follows at 0.85 against 0.162; A3 tuning at 0.8 against 0.116. The inversion is the whole point: the map that is provably empty on the position regime looks *more* convincing than the map that is exactly right. We state the obvious caveat plainly. The plausibility axis is a documented method-tradition proxy, not a human-subjects measurement, and we report it as such to avoid overclaiming the danger zone. Yet the faithfulness axis is exact, and the ranking it inverts is the field’s own habit of trust; the hazard is therefore real even at the proxy’s coarse resolution. The danger zone is not a worry on the VCS. It is a measurement.

Where the leaderboard stops — the semantic gap

Here the paper turns. Suppose we hand the audit its best case: a method that scores 1.0 against the oracle, a faithfulness no real network can certify. Has it given us an understanding of what the program does? We claim, and now show, that it has not—and we are careful about what that claim is and is not. We do *not* claim that no method can recover meaning; that is the unbounded statement, and it is the subject of the companion design-recovery work (Paper 4), which we have not run here. We claim the bounded thing the oracle can prove: even when a method recovers the true causal structure exactly, what it returns is the *wiring* —a causal-effect map, a data-flow graph—and not the *semantics*: the variable’s meaning or the routine’s algorithm. The missing object is of a different type, not a worse-fit instance of the same one. The triad of Sec. 3.4 makes this operational: an explanation is right only if it is **Faithful** (true causes), **Sufficient** (it regenerates the behavior under held-out interventions), and **Minimal** (at the algorithmic level). Three committed separations, each adjudicated by the oracle rather than by our preference, show the axes pulling apart—and, tellingly, pulling apart in *both directions at once* (Fig. 5).

Graph-correct, behavior-incomplete.

Automatic circuit discovery (ACDC) [24] on Breakout recovers a data-flow graph that is, by the exact data-flow oracle, perfectly correct and perfectly parsimonious: $F = 1.0$ and $M = 1.0$ [34]. And yet the discovered circuit alone, with its non-circuit parents scrubbed, reproduces the clean candidate-cell readouts only 44% of the time: $S = 0.44$. A correct graph of which cells feed which is not the function those cells compute. The companion causal method shows the same thing from the faithful side: activation patching [21] recovers the exact per-site causal-effect table on every one of the six games—the record carries `recovered_equals_exact: true` bit-for-bit [34]—and a

table of effects, however exact, is not the bounce algorithm that produced them. Crucially, there is no tuning budget left to spend here: the effect table is *already* exact, so the residual to the algorithm cannot be a worse fit closed by a better hyperparameter. It is a missing object of a different type. This is the sharpest reading of the triad and the reason the gap is not merely “methods underperform.”

Named, yet causally inert.

The reverse failure is just as instructive. A sparse autoencoder [27] trained on the recorded Pong state matches every one of its candidate game-concept variables: the feature $\leftrightarrow$ variable matched fraction is 1.0, a perfect semantic alignment by the field’s usual yardstick. By the oracle, those same matched features are nearly causally inert: faithfulness $F = 0.041$ and a *negative* sufficiency $S = -0.336$ [34]. “Looks like the ball’s x ” is not “is how the program computes with the ball’s x .” Where the circuit was correct yet incomplete, the dictionary is named yet wrong—the gap runs the other way on the same axis, and a story in which methods simply score too low cannot produce both at once. We add the obvious guard against reading semantic success into an aligned variable: the interchange-intervention alignment (DAS) [26] reports an aligned-variable accuracy of 1.0, but that 1.0 is true *by construction*—it verifies a *supplied* variable rather than discovering one. A method that confirms a given concept is not a method that found the meaning.

Decodable, yet unused.

The third separation is the present-versus-used trap, and the oracle is strictly stronger than the probe that walks into it. Linear probing with control tasks [8] decodes game concepts from the Breakout state at a mean accuracy of 0.72 with positive selectivity over its control—the information is demonstrably *present*. Yet the exact oracle flags a decodable score-related cell (cell 84) as *not causally used*: re-derived within the horizon, its value has no downstream effect on the output [34]. A concept can be linearly readable from the state and play no part in the computation, exactly Hewitt and Liang’s control-task warning—decodability is necessary, never sufficient. Probing answers “is it there”; the oracle answers “is it used,” and only the second is about how the program works. Linear probing’s leaderboard faithfulness of 0.162 paired with a plausibility proxy of 0.85 is its place in the danger zone, and the decodable-but-unused cell is the mechanism behind that placement.

The gap is a number, and a different kind of object.

Read together (the leaderboard plane of Fig. 4 and the failure taxonomy of Fig. 5), the three separations make the same point from three sides. The triad’s axes are not redundant: a method can be faithful and minimal yet not sufficient (ACDC, $F = M = 1.0$, $S = 0.44$); it can match a named concept perfectly yet be causally inert (the SAE, matched fraction 1.0, $F = 0.041$); it can decode a variable that the program never uses (cell 84). The robustness of the claim does not rest on any single low number—a lone low S could be a tuning artifact—but on the gap appearing where tuning *cannot* reach (an already-exact effect table) and running in opposite directions on the same axis at once. That two-sidedness is what a “methods are not good enough yet” account

cannot generate. We restate the honesty contract here, because the temptation to over-read is greatest at the paper’s centre. We have *not* recovered the program’s meaning, algorithm, variables, or design from scratch. Every semantic label we used is external—imported from AtariARI [6] and OCArari [7], `verified:false` in provenance—and the substrate only *causally verifies* a supplied label; it does not *discover* the concept. The faithful methods recover the wiring; the meaning is a separate object, and the distance to it is what we have measured. Recovering that meaning is a different problem, and it is the companion paper’s, not this one’s.

8 Discussion

In this Article, we have turned the interpretability toolkit onto a machine whose ground truth is computable, and we could measure two things at once. We have shown, first, that on a system where the true cause of any output is recoverable by exact intervention, causal and intervention methods are faithful while the field’s most familiar gradient and correlational tools are not—the leaderboard separates them along the only axis that admits an exact answer (Fig. 4). We have shown, second, that this is the easier half of the problem. Even a method that scores at the oracle ceiling returns the causal *wiring*—an effect map, a data-flow graph—and not the *semantics*, the variable’s meaning or the routine’s algorithm. The residual between the two is not a matter of taste. On this machine it is a number, and we have named it the *semantic gap*. We discuss here what that gap implies for interpretability practice, why a 1977 chip is a fair place to measure it, where our claim stops, and what would be needed to close it.

Many traditions, one ground truth — and the boundary it cannot cross

The first lesson is a confirmation, made quantitative. Three research traditions that rarely share a page—a neuroscience battery, the attribution toolkit, and mechanistic interpretability—turn out to occupy a single plane once a common oracle scores them (Fig. 4). Jonas and Kording warned that an analyst armed with the methods of systems neuroscience could fail to understand even a microprocessor, because finding structure is not the same as explaining it [2]. We have made their lesson a measurement: the classical battery recovers rich structure yet scores low against the known mechanism, the gradient family collapses to the floor on the discrete position regime, and only the causal operators reach the ceiling. Mechanistic interpretability, in particular, has until now been calibrated against *synthetic* or *compiled* ground truth—programs compiled into transformers by construction, or semi-synthetic networks with a planted circuit [36, 37]; ground-truth *attribution* benchmarks likewise plant the important features [53]. We add the missing case: a real, hand-authored, deployed artifact, never designed to be interpretable, whose ground truth is *independent* of the artifact rather than built into it. *Hence* the leaderboard is not only a ranking; it is the first ground-truth calibration of these methods on a system the field did not get to plant.

Yet the same oracle marks a boundary the leaderboard cannot cross, and that boundary is the real subject of this paper. A perfect faithfulness score certifies that

a method named the true causes. It does not certify that the method recovered what the program *means*. We saw the boundary three times, from three sides (Fig. 5): a circuit that is graph-correct and minimal ($F = M = 1.0$) yet reproduces behavior only 44% of the time ($S = 0.44$); a feature dictionary that matches every named variable (matched fraction 1.0) while being nearly causally inert ($F = 0.041$, $S = -0.336$); and a concept that is linearly decodable yet, by the oracle, never used in the computation. The gap runs in both directions on the same axis at once—correct-but-incomplete and named-but-wrong— which is precisely what a “methods are not good enough yet” story cannot produce.

What *understanding* would require — the semantic level

Why does faithfulness stop short? The cleanest answer comes from Marr’s levels of analysis [40]. A faithful attribution lives at the implementational level: it tells us which cells and registers carry the signal and how a perturbation propagates. Understanding lives one level up, at the algorithmic level—what function the program computes and how. A correct data-flow graph is an implementational fact; the bounce algorithm it implements is an algorithmic one, and the two are objects of a different type. This is also Lazebnik’s lament, that one can characterize a system in exhaustive mechanistic detail and still not understand it [54]. Our contribution is to turn that lament into a measured quantity: the distance from a faithful effect table to the algorithm is the sufficiency gap $1 - S$, and on the exact effect table that gap cannot be closed by a better hyperparameter, because the table is *already* exact.

The sharpest way to state the bar is to borrow it from software engineering. IEEE 610.12 defines software as programs *plus* their documentation—the specification, the design, the intent [55]. A faithful circuit recovers part of the program. Understanding the system means recovering its documentation: what each variable is *for*, what each routine is *meant* to do, the design the engineer held in mind. Our causal oracle, by construction, cannot reach that object, because the meaning of a variable is not a causal fact about the variable—it is a fact about the design that the variable serves. *Hence* the gap we measure is not an artifact of weak methods but a property of the question: faithfulness is necessary for understanding, and it is not sufficient. We are deliberate that this paper *defines and measures* the gap. *Closing* it —recovering the program’s documentation and design from the bare machine—is the reverse-engineering bar, and it is the companion design-recovery paper (Paper 4), which frames itself as measuring the distance to that bar, not clearing it. The behavioral-grounding bridge—asking whether a recovered concept matches how the system is *used*—is the companion behavior paper (Paper 3). The learned agent, where there is no disassembly to check against, is Paper 5.

A second strand of the literature reads our result the other way around. Probing shows that a concept is *present* in the state; it does not show that the concept is *used*, the control-task warning of Hewitt and Liang [8]. Our intervention oracle is strictly stronger: it asks whether the byte *causes* the output, and it found a decodable score cell that the program never reads. *Note that* this is the same boundary seen from the data side. Decodability is an implementational presence; use is an algorithmic fact; understanding needs the second, and only the oracle supplies it.

The shared toolkit, read across disciplines

It is worth saying plainly that the three traditions are, at bottom, one toolkit under different names. Tuning curves in neuroscience are feature visualization in mechanistic interpretability; lesions are ablations; connectomics is circuit discovery; dimensionality reduction is probing’s cousin; Granger causality is the correlational ancestor of causal attribution. Each pair shares a method and a failure mode. *Hence* a result measured on one transfers as a caution to the others: the tuning-curve trap and the probing trap are the same trap, and the danger zone of Fig. 4 is populated by both traditions at once. This cross-disciplinary reach is the reason a benchmark built on an Atari chip speaks to systems neuroscience, to explainable AI, to mechanistic interpretability, and to software verification in the same breath.

Is a 1977 chip a fair test? The necessary-condition screen

The obvious objection deserves a full hearing. The VCS is deterministic, hand-coded, and small; the systems these methods are built for are learned, distributed, stochastic, and large. Why should a score here predict anything about a transformer? Our answer is that we do not ask it to. The benchmark is a *necessary-condition screen*, not a sufficiency predictor. A method that fails *here*—with perfect ground truth, full observability, exact interventions, and no learning to muddy the picture—has no business being trusted on a neural network, where every one of those advantages is gone. Passing here is necessary, not sufficient; failing here is decisive.

In fact, the same logic upgrades the semantic-gap claim, and this is the part worth stating carefully. If even the *strongest* causal account that a fully observable machine permits stops short of meaning, then on a neural network—less observable, with no clean ground truth—faithfulness is *a fortiori* not understanding. The gap we measure here is therefore a floor on the gap there. We measure $1 - S = 0.56$ on an exactly recovered circuit; we should not expect to do better where the circuit itself is uncertain.

The objection also has a constructive answer: the VCS *does* exhibit the properties that make real systems hard to interpret. Its RAM cells are aliased and polysemantic, reused across routines, a hand-coded echo of superposition; it computes by racing the electron beam, a distributed temporal computation; it reads a floating bus; its registers mean different things in different contexts. Figure 6 maps each VCS property to the neural-network difficulty it does and does not capture—a property provable here is a property suspected there. What the chip does *not* capture we state without hedging: learned features, scale, and stochasticity are absent, and the learned agent, which folds all three back in, is Paper 5. The map is the honest accounting of where the analogy holds and where it breaks.

What interpretability needs

Three recommendations follow, and we state them flatly. *First*, prefer causal over correlational evidence. The danger zone is not a hypothetical: the methods the field reaches for first carry the highest plausibility proxy and the lowest measured faithfulness, and a map that is provably empty on the position regime looks *more* convincing

than the map that is exactly right. *Second*, validate on known systems before trusting on unknown ones—sieve the methods through a ground-truth screen, because a method that cannot pass where the answer is computable cannot be audited where it is not. *Third*, and most important, stop treating faithfulness as understanding. The open problem for explainable AI is not a better saliency map; it is the semantic gap. Faithfulness is the half we now know how to measure and, on causal methods, how to win. Meaning is the half that remains.

Limitations

We have measured a gap; we have not closed it, and several limits bound exactly what we may conclude. *First*, we did not run semantic recovery. Phase E (documentation and design recovery) is specified but not executed, so our claim is the *bounded* one—on exact ground truth, faithfulness certifies the wiring, and the wiring alone is not the semantics—and *not* the unbounded “no method can recover meaning,” which would require running the recovery and is Paper 4’s charge. *Second*, the sufficiency and minimality axes are causal quantities defined against the oracle, and the T3 semantic labels are *external*, imported from AtariARI [6] and OCArari [7] and only causally verified, never discovered. We say this plainly, and we turn it into the point: even the strongest causal account a fully known machine permits stops short of meaning, and recovering meaning from scratch is a different problem. *Third*, a single low sufficiency could be a tuning artifact. The claim therefore does not rest on any one number; it rests on the gap appearing where tuning cannot reach—an already-exact effect table—and on its running in both directions on the same axis at once. *Fourth*, the plausibility axis is a documented method-tradition proxy, not a human-subjects measurement, and we report the danger zone at that proxy’s coarse resolution rather than overclaiming it. *Fifth*, our scores hold within Paper 1’s validated conformance window; outside that horizon the substrate is not certified bit-exact, and we keep every scored claim inside it. *Finally*, the subject is one old machine with no learning. That is the necessary-condition framing again, conceded once more: a floor, not a forecast.

Beyond the brain, and toward the learned agent

We close by reaching past the present result. Because the substrate is differentiable, it tells us which *gradient* explanations are trustworthy and where they vanish—the content path carries an honest gradient, while the discrete index does not, so a saliency map over sprite position is empty by construction, not by accident. That is a statement about gradients as a class, and it travels to any system where a discrete decision sits downstream of a continuous one. We believe the broader value of a fully known, differentiable testbed is exactly this: it lets the community calibrate its instruments before deploying them where calibration is impossible. The hard case remains the learned agent, where there is no disassembly, no specification, no engineer’s intent to recover, and where the semantic gap we measured here can only widen. It is foreseeable that closing that gap—turning a faithful account of a learned system into an understanding of what it has learned—will be the work of the next decade of interpretability. We

have not solved it. We have made it a number, and a number is where understanding begins.

Data availability. The interpretability scores reported here are derived from the committed analysis records under `tools/xai_study/` (the per-method, per-game §R records and the aggregated cross-tradition leaderboard [33, 34]); every quantitative claim in the Article reads from one of these committed artifacts, and no number is re-run or estimated. The game cartridges used as the subject are third-party Atari 2600 ROMs that we do *not* redistribute; following the practice of the underlying emulator port [4], we provide the SHA-256 hashes of each ROM and the `AutoROM` retrieval procedure so that the exact binaries can be reproduced independently. The recorded state trajectories used to train the sparse autoencoders and to run the patching and probing experiments are regenerated deterministically from those ROMs and the published action streams.

Code availability. The differentiable, bit-exact emulator in its two interchangeable realizations (`jutari`, Julia; `jaxtari`, JAX/Python) and the ground-truth intervention oracle are described in the companion port paper [4]. The interpretability benchmark introduced here—the scoring suite, the $F \wedge S \wedge M$ triad metrics, and the per-phase method implementations (`tools/xai_study/`)—will be released as a reusable open-source artifact. The full source for the emulator, the oracle, and the benchmark will be made available under the MIT license upon acceptance.

Author contributions. *[CRediT placeholder, to be finalised with the author list.]* A.M. conceived the study, designed the ground-truth audit and the $F \wedge S \wedge M$ triad, and wrote the manuscript. S.B. and P.K. contributed to the method design, the substrate and oracle, and the analysis. All authors discussed the results, interpreted the semantic-gap finding, and revised the manuscript. The differentiability framing and the gradient-faithfulness analysis were developed jointly with P.K.

Competing interests. The authors declare no competing interests.

Ethics and broader impact. This study uses no human-subjects data; the human-plausibility axis is a documented method-tradition proxy rather than a measurement, and we report it as such. The subject is a fully deterministic, hand-coded program with no learned policy, so no agent is trained. Interpretability methods are dual-use, in that a tool that audits a system can also be used to probe it; our contribution is a calibration that makes explanation claims more, not less, trustworthy, and we release it openly to that end. The game cartridges are not redistributed.

Acknowledgements. *[To be completed: funding sources and compute allocations.]* The authors thank the maintainers of the AtariARI [6] and OCAAtari [7] projects, whose game-concept labels we imported as external candidates and only causally verified, and the maintainers of the `xitari` reference emulator [5] against which the substrate is validated bit-for-bit. The pilots were run locally; the full sweeps used the institutional compute cluster.

References

- [1] Greydanus, S., Koul, A., Dodge, J. & Fern, A. *Visualizing and understanding Atari agents. Proceedings of the 35th International Conference on Machine Learning (ICML)*, Vol. 80, 1792–1801 (PMLR, 2018).
- [2] Jonas, E. & Kording, K. P. Could a neuroscientist understand a microprocessor? *PLOS Computational Biology* **13**, e1005268 (2017).
- [3] Atrey, A., Clary, K. & Jensen, D. *Exploratory not explanatory: Counterfactual analysis of saliency maps for deep reinforcement learning. International Conference on Learning Representations (ICLR)* (2020).
- [4] Maier, A., Bayer, S. & Krauss, P. A bit-exact, differentiable atari 2600 (paper 1) (2025). Companion paper; this program.
- [5] DeepMind. Xitari: An Arcade Learning Environment fork. <https://github.com/google-deepmind/xitari> (2016). Accessed: 2026-06-16.
- [6] Anand, A. *et al.* *Unsupervised state representation learning in atari. Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32 (2019).
- [7] Delfosse, Q., Blüml, J., Gregori, B., Sztwiertnia, S. & Kersting, K. OCArari: Object-centric Atari 2600 reinforcement learning environments (2023). [2306.08649](#).
- [8] Hewitt, J. & Liang, P. *Designing and interpreting probes with control tasks. Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2733–2743 (2019).
- [9] Simonyan, K., Vedaldi, A. & Zisserman, A. *Deep inside convolutional networks: Visualising image classification models and saliency maps. International Conference on Learning Representations (ICLR) Workshop* (2014).
- [10] Sundararajan, M., Taly, A. & Yan, Q. *Axiomatic attribution for deep networks. Proceedings of the 34th International Conference on Machine Learning (ICML)*, Vol. 70, 3319–3328 (PMLR, 2017).
- [11] Smilkov, D., Thorat, N., Kim, B., Viégas, F. & Wattenberg, M. *Smoothgrad: Removing noise by adding noise. ICML Workshop on Visualization for Deep Learning* (2017).
- [12] Springenberg, J. T., Dosovitskiy, A., Brox, T. & Riedmiller, M. *Striving for simplicity: The all convolutional net. International Conference on Learning Representations (ICLR) Workshop* (2015).

- [13] Selvaraju, R. R. *et al.* Grad-CAM: Visual explanations from deep networks via gradient-based localization. *International Journal of Computer Vision* **128**, 336–359 (2020). Originally arXiv:1610.02391 (2016).
- [14] Chattopadhyay, A., Sarkar, A., Howlader, P. & Balasubramanian, V. N. *Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks*. *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 839–847 (IEEE, 2018).
- [15] Zeiler, M. D. & Fergus, R. *Visualizing and understanding convolutional networks*. *European Conference on Computer Vision (ECCV)*, 818–833 (2014).
- [16] Fong, R. C. & Vedaldi, A. *Interpretable explanations of black boxes by meaningful perturbation*. *IEEE International Conference on Computer Vision (ICCV)*, 3429–3437 (2017).
- [17] Petsiuk, V., Das, A. & Saenko, K. *RISE: Randomized input sampling for explanation of black-box models*. *British Machine Vision Conference (BMVC)* (2018).
- [18] Ribeiro, M. T., Singh, S. & Guestrin, C. “*why should i trust you?*”: Explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1135–1144 (2016).
- [19] Lundberg, S. M. & Lee, S.-I. *A unified approach to interpreting model predictions*. *Advances in Neural Information Processing Systems (NeurIPS)* (2017).
- [20] Such, F. P. *et al.* *An Atari model zoo for analyzing, visualizing, and comparing deep reinforcement learning agents*. *Proceedings of the 28th International Joint Conference on Artificial Intelligence (IJCAI)*, 3260–3267 (2019).
- [21] Vig, J. *et al.* *Investigating gender bias in language models using causal mediation analysis*. *Advances in Neural Information Processing Systems (NeurIPS)* (2020).
- [22] Meng, K., Bau, D., Andonian, A. & Belinkov, Y. *Locating and editing factual associations in GPT*. *Advances in Neural Information Processing Systems (NeurIPS)* (2022).
- [23] Wang, K., Variengien, A., Conmy, A., Shlegeris, B. & Steinhardt, J. *Interpretability in the wild: A circuit for indirect object identification in GPT-2 small*. *International Conference on Learning Representations (ICLR)* (2023).
- [24] Conmy, A., Mavor-Parker, A. N., Lynch, A., Heimersheim, S. & Garriga-Alonso, A. *Towards automated circuit discovery for mechanistic interpretability*. *Advances in Neural Information Processing Systems (NeurIPS)* (2023).

- [25] Geiger, A., Lu, H., Icard, T. & Potts, C. *Causal abstractions of neural networks*. *Advances in Neural Information Processing Systems (NeurIPS)* (2021).
- [26] Geiger, A., Wu, Z., Potts, C., Icard, T. & Goodman, N. D. *Finding alignments between interpretable causal variables and distributed neural representations*. *Proceedings of the Third Conference on Causal Learning and Reasoning (CLEaR)* (2024).
- [27] Cunningham, H., Ewart, A., Riggs, L., Huben, R. & Sharkey, L. Sparse autoencoders find highly interpretable features in language models (2023). [2309.08600](#).
- [28] Alain, G. & Bengio, Y. *Understanding intermediate layers using linear classifier probes*. *International Conference on Learning Representations (ICLR) Workshop* (2017).
- [29] Belrose, N. *et al.* Eliciting latent predictions from transformers with the tuned lens (2023). [2303.08112](#).
- [30] Nanda, N. Attribution patching: Activation patching at industrial scale. <https://www.neelnanda.io/mechanistic-interpretability/attribution-patching> (2023).
- [31] Jacovi, A. & Goldberg, Y. *Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness?* *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 4198–4205 (Association for Computational Linguistics, 2020).
- [32] Adebayo, J. *et al.* *Sanity checks for saliency maps*. *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 31, 9505–9515 (2018).
- [33] Paper 2 authors. Faithful-method demonstration: causal near-ceiling versus popular near-chance on the differentiable Atari VCS. Committed analysis record, `tools/xai_study/compare/out/faithful_demo.md`; pure read over the committed cross-tradition leaderboard (`compare/out/leaderboard.json`) and the per-game §R records it aggregates. Generated by `compare/faithful_demo.py`. (2026). Internal data artifact. Position-regime pair `activation_patching = 1.000` vs `vanilla_saliency = 0.000` on all six core games (per-method gap 1.000). Accessed 2026-06-23.
- [34] Paper 2 authors. Cross-tradition faithfulness-plausibility leaderboard for the differentiable Atari VCS interpretability audit (phases a–c). Committed analysis record, `tools/xai_study/compare/out/leaderboard.json`; aggregates the per-game §R records under `tools/xai_study/phaseA_kording/out/` (2026). Internal data artifact; 257 per-game records aggregated into 31 method rows including the oracle positive control. Generated by `compare/leaderboard.py`. Accessed 2026-06-23.

- [35] Yang, M. & Kim, B. *Benchmarking attribution methods with relative feature importance*. *NeurIPS Workshop / arXiv:1907.09701* (2019). Introduces the BAM benchmark with known relative feature importance and false-positive metrics for attribution methods., [1907.09701](#).
- [36] Lindner, D. *et al.* *Tracr: Compiled transformers as a laboratory for interpretability*. *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 36 (2023). [2301.05062](#).
- [37] Gupta, R., Arcuschin, I., Kwa, T. & Garriga-Alonso, A. *InterpBench: Semi-synthetic transformers for evaluating mechanistic interpretability techniques*. *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, Vol. 37 (2024). [2407.14494](#).
- [38] Bellemare, M. G., Naddaf, Y., Veness, J. & Bowling, M. The Arcade Learning Environment: An evaluation platform for general agents. *Journal of Artificial Intelligence Research* **47**, 253–279 (2013).
- [39] Barbiero, P. *et al.* Neural interpretable reasoning. *arXiv preprint arXiv:2502.11639* (2025).
- [40] Marr, D. *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information* (W. H. Freeman and Company, 1982).
- [41] Lazebnik, Y. Can a biologist fix a radio? — or, what i learned while studying apoptosis. *Cancer Cell* **2**, 179–182 (2002).
- [42] Bengio, Y., Léonard, N. & Courville, A. Estimating or propagating gradients through stochastic neurons for conditional computation (2013). [1308.3432](#).
- [43] Jang, E., Gu, S. & Poole, B. *Categorical reparameterization with Gumbel-softmax*. *International Conference on Learning Representations (ICLR)* (2017).
- [44] Jaderberg, M., Simonyan, K., Zisserman, A. & Kavukcuoglu, K. *Spatial transformer networks*. *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 28 (2015).
- [45] Abnar, S. & Zuidema, W. *Quantifying attention flow in transformers*. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 4190–4197 (2020).
- [46] Chan, L. *et al.* Causal scrubbing: A method for rigorously testing interpretability hypotheses. *AI Alignment Forum* (2022).
- [47] Shrikumar, A., Greenside, P. & Kundaje, A. *Learning important features through propagating activation differences*. *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 3145–3153 (2017).

- [48] Erion, G., Janizek, J. D., Sturmfels, P., Lundberg, S. M. & Lee, S.-I. Improving performance of deep learning models with axiomatic attribution priors and expected gradients. *Nature Machine Intelligence* **3**, 620–631 (2021).
- [49] Štrumbelj, E. & Kononenko, I. Explaining prediction models and individual predictions with feature contributions. *Knowledge and Information Systems* **41**, 647–665 (2014).
- [50] Fong, R., Patrick, M. & Vedaldi, A. *Understanding deep networks via extremal perturbations and smooth masks*. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2950–2958 (2019).
- [51] Olson, M. L., Khanna, R., Neal, L., Li, F. & Wong, W.-K. Counterfactual state explanations for reinforcement learning agents via generative deep learning. *Artificial Intelligence* **295**, 103455 (2021).
- [52] Bastani, O., Pu, Y. & Solar-Lezama, A. *Verifiable reinforcement learning via policy extraction*. *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 31, 2494–2504 (2018).
- [53] Yang, M. & Kim, B. BIM: Towards quantitative evaluation of interpretability methods with ground truth. arXiv preprint (2019). ArXiv:1907.09701.
- [54] Lazebnik, Y. Can a biologist fix a radio? — or, what i learned while studying apoptosis. *Cancer Cell* **2**, 179–182 (2002).
- [55] IEEE. IEEE standard glossary of software engineering terminology (ieee std 610.12-1990). IEEE, New York (1990). Defines “software” as computer programs, procedures, and possibly associated documentation and data.

Figure 6 — A taxonomy of interpretability failure modes on the VCS, by underlying mechanism

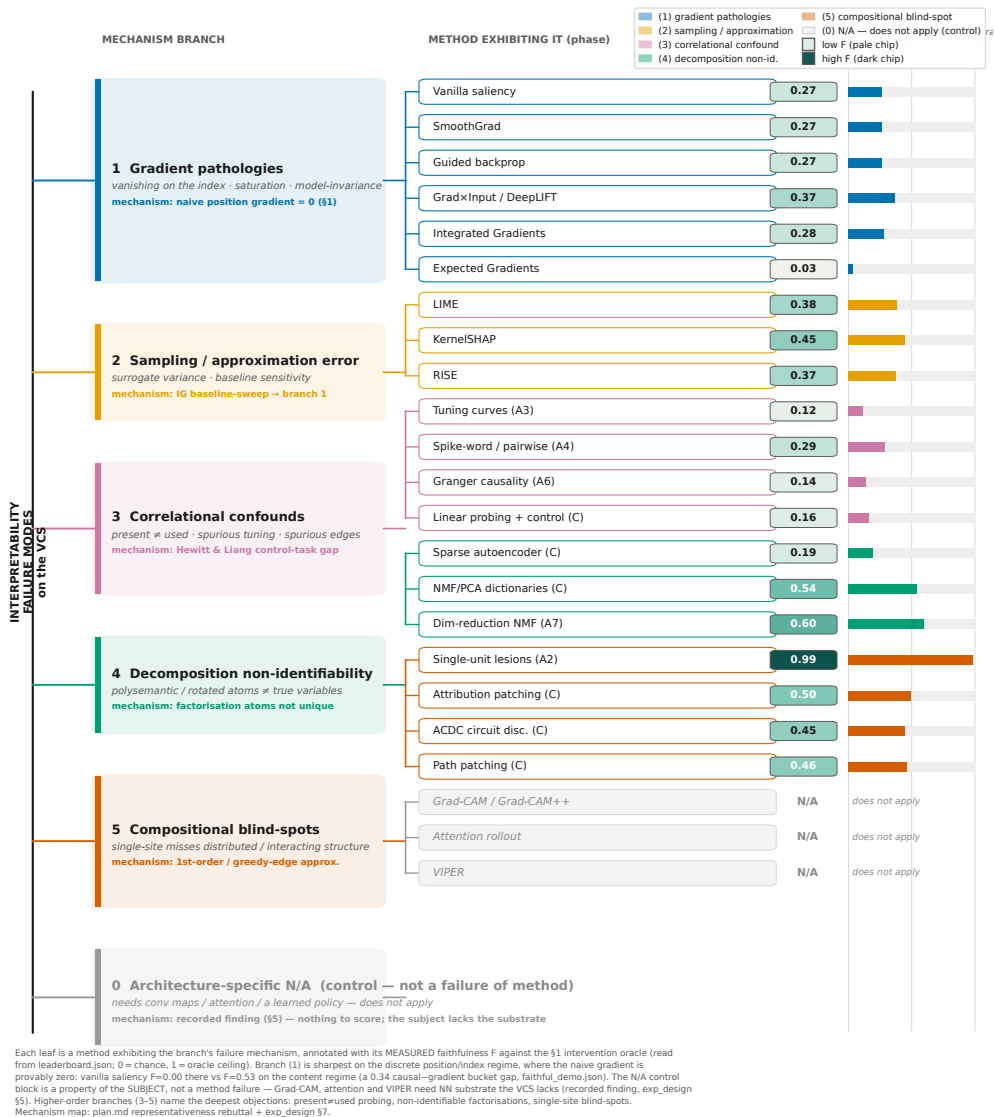


Fig. 5 A taxonomy of interpretability failure on the VCS, organised by underlying mechanism — and the boundary the leaderboard cannot cross. Each leaf is a method exhibiting the branch's failure mode, annotated with its measured faithfulness F against the §3.2 oracle (read from the committed leaderboard; 0 = chance, 1 = oracle ceiling). The lower branches are the *confirm* half: the gradient family is provably zero on the discrete position/index regime, and the architecture-specific block (Grad-CAM, attention, VIPER) does not apply because the VCS lacks the neural substrate they require — a property of the subject, not a method failure. *Note that* the higher-order branches are the *semantic gap*: present-but-unused concepts (linear probing flags a decodable score cell the oracle says is causally inert), non-identifiable factorisations (the SAE matches every named variable at fraction 1.0 yet scores $F = 0.041$), and graph-correct-but-behavior-incomplete circuits (ACDC at $F = M = 1.0$, $S = 0.44$). These are the failures that survive a perfect faithfulness score, because the missing object — the variable's meaning, the routine's algorithm — is of a different type. All semantic labels are external (AtariARI/OCAtari) and only causally verified, never discovered. Every annotated value reads from a committed record.

The representativeness map: every failure mode we measure on the VCS has a documented neural-network counterpart

The single strongest objection — “a deterministic 1977 chip cannot stand in for a learned, distributed, stochastic neural network” — answered as a necessary-condition screen.

The asymmetry is the argument: on the VCS each failure is **PROVABLE** — exact intervention oracle, full observability, no learning; on neural networks the same failure is only **SUSPECTED**. A method that fails here has no business being trusted there (necessary, not sufficient).

Failure mode observed & MEASURED on the VCS	VCS evidence (PROVABLE) scored vs the exact \$1 oracle	Neural-network counterpart documented analogue (cite-ready)	Status here → there
1 Gradient vanishes on discrete / index outputs	0.000 position-regime F Vanilla saliency & IG faithfulness = 0.000 on the position regime; whole gradient bucket = 0.068 (n=19). Provably 0 (strobe-timed argmax, \$1).	Shattered / saturated gradients; non-differentiable argmax & index readouts. [Balduzzi 2017; Sunderarajan 2017 (IG motivation)]	PROVABLE here ↓ suspected there
2 Saliency is model-invariant (Adebayo sanity-check fails)	0.000 sanity-check margin Program-randomization test: saliency cosine = 1.000 while 95% of RAM changed → sanity check FAILS. Constant jacobian, no ReLU.	Saliency unchanged under model- & label-randomization; looks like an edge detector. [Adebayo et al. 2018; NeurIPS]	PROVABLE here ↓ suspected there
3 Correlation ≠ causation (Granger & tuning trap)	0.136 Granger F Granger edges faithfulness = 0.136 (false edges vs the true data-flow); game-variable tuning = 0.116 (spurious tuning) — vs exact oracle.	Spurious Granger edges; “present ≠ used” tuning & attention as explanation. [Anand 2019; Jain & Wallace 2019 (attention)]	PROVABLE here ↓ suspected there
4 Single-unit ablation has an interaction blind-spot	0.414 connectome F Single-unit lesion importance is faithful (0.987) yet MISSES the wiring: connectome graph recovered at only 0.414. SAE feature ablations move y in 0 of the audited cells.	Single-unit ablation misleading: computation is distributed across units. [Morcos et al. 2018; Leavitt & Morcos 2020]	PROVABLE here ↓ suspected there
5 SAE / probe: present-not-used + polysemanticity	0.162 selectivity F SAEs match a known var (100% of candidates in 4/8 games, min 83%) yet score 0.195 on causal use; probes decode at 0.894 acc but 25 cells are decodable-not-causal → selectivity F = 0.162.	Control-task selectivity gap: superposition / polysemantic neurons (features in superposition). [Hewitt & Liang 2019; Elhage et al. 2022]	PROVABLE here ↓ suspected there
6 Baseline-sensitivity of Integrated / Expected Gradients	0.034 EG F (sh-regime) IG magnitude is baseline-sensitive (corr_sensitivity up to 0.999 across games); a baseline = the content byte kills IG; EG lands at 0.034.	Attribution depends on the (arbitrary) baseline choice; no canonical reference input. [Kondemans et al. 2019; Sturmfels et al. 2020]	PROVABLE here ↓ suspected there

Headline (position regime): causal/intervention 0.412 vs gradient/correlational 0.068 — a 0.344 faithfulness gap; activation patching = 1.000 (oracle ceiling) where vanilla saliency = 0.000 (naive-gradient floor).

■ VCS evidence — measured vs the exact \$1 intervention oracle ■ PROVABLE on the VCS (ground truth)
■ NN counterpart — documented analogue (conceptual mapping, cite-ready) ■ only SUSPECTED on neural nets

All VCS numbers read from committed records: leaderboard.json (P2-E8-1); faithful_demo.json (P2-E8-3); guided_backprop_core_summary (Adebayo); ig_baseline_sweep_core_summary; sae_core_summary; linear_probing_core_summary. No experiment re-run. NN-counterpart labels per experiment_design.md §4-7 and plan.md (the representativeness rebuttal).

Fig. 6 The VCS↔neural-network failure-mode map — why a 1977 chip is a fair screen. Each row is an interpretability failure mode we *measured* on the VCS (left, blue), paired with the well-documented neural-network counterpart it stands in for (right, vermillion). The status column carries the necessary-condition argument: on the VCS each failure is *provable* because we hold the exact ground truth, whereas on a neural network the same failure is only *suspected*. Among the rows: vanilla saliency and integrated gradients score 0.000 on the discrete position regime (the shattered-gradient analogue); Granger causality reaches faithfulness 0.136 with a high false-edge rate (the spurious-edge analogue); the single-unit lesion view is faithful (0.987) yet *misses* the wiring that connectomics recovers (0.41); and the sparse autoencoder matches every named variable yet scores $F = 0.041$ (the present-not-used and superposition analogue). *Note that* passing this screen is necessary, not sufficient: a method that fails here—with perfect ground truth, full observability, exact interventions, and no learning—has no business being trusted where every one of those advantages is gone. What the VCS does *not* capture (learned features, scale, stochasticity) is stated in the legend and deferred to Paper 5. Every VCS value reads from a committed record.